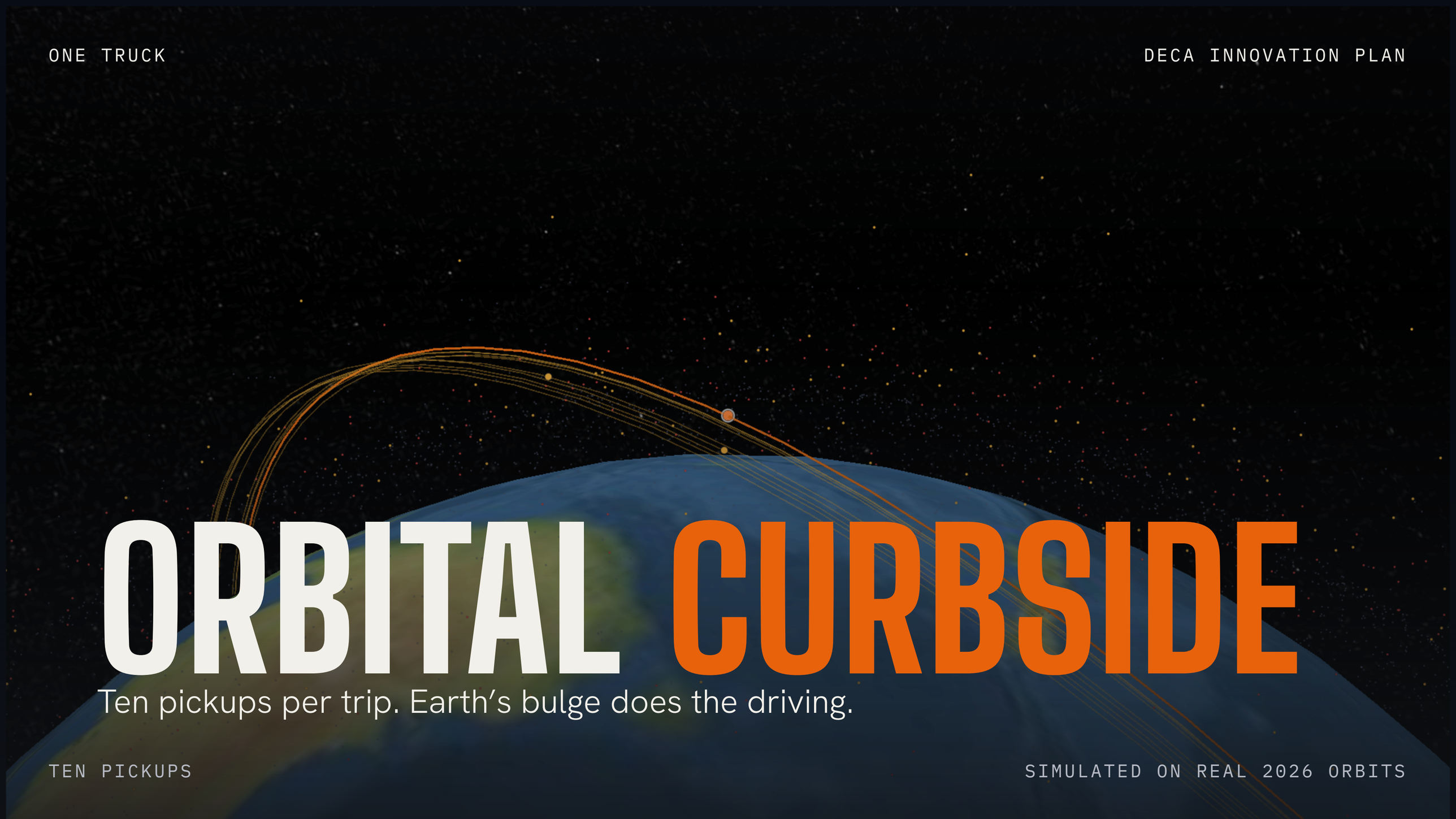


ORBITAL CURBSIDE

Ten pickups per trip. Earth’s bulge does the driving.

INNOVATION PLAN

Orbital Curbside

Removing dead satellites and rocket stages ten at a time, priced like a garbage route.



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[Presentation date]

DECA INNOVATION PLAN (EIP) · 2026–27

Contents

I Overview 4

II Business Opportunity 5

III Customer Segments 8

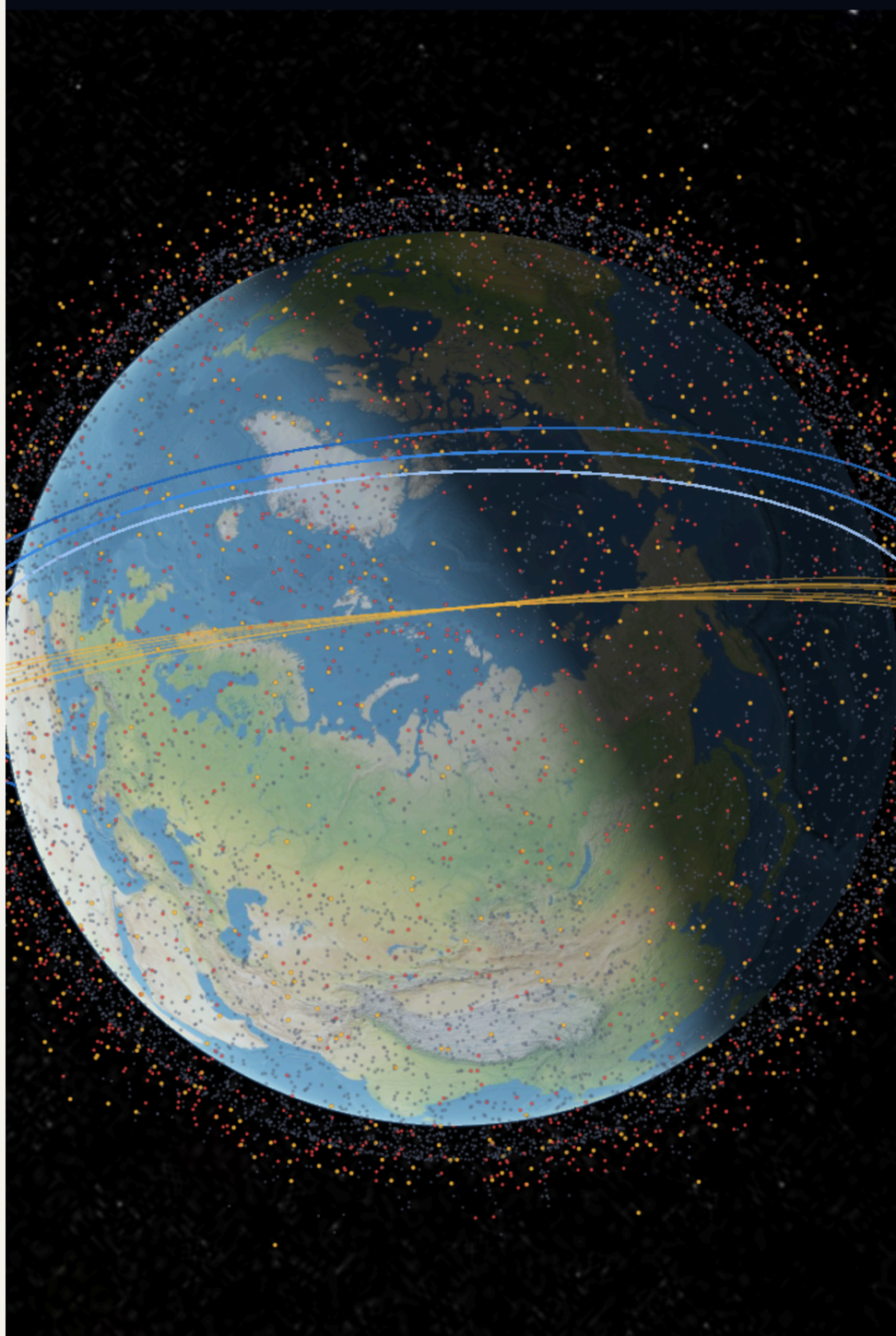
IV Unique Value Proposition 11

V Solution 13

VI Conclusion 18

VII Bibliography 19

VIII Appendix 20



One electric spacecraft removes ten dead objects per trip, by letting Earth’s bulge do the driving.

ABOUT

A space “garbage truck” that parks, drifts, hops and tags each object with a deorbit kit.

BUSINESS MODEL

Removal as a service: a fixed fee per object, paid on confirmed deorbit.

CUSTOMERS

1 Space agencies
2 Satellite operators
3 Space insurers

VALUE

$12.2M

modelled per removal, launch included, vs $40–93M per object today

About 47,000 objects are tracked in orbit. ESA says the junk keeps multiplying even if we never launch again.



Our simulation, real catalogue of 16 Oct 2026 · ESA DISCOSweb (Oct 2026); ESA Space Environment Report 2026

Today, removing one dead object is a custom mission costing $40–93 million.

Governments have started paying for removal, but they buy one or two objects at a time.

Mission (buyer)	Objects	Public value
ClearSpace-1 (ESA, 2020)	1	€86M
ADRAS-J2 (JAXA, 2024)	1	≈¥13.2B (≈$82M)
UK ADR (UK Space Agency, Oct 2026)	2	≈£61M (≈$80M)
Disposal service (US SDA, Jan 2026)	1 + options	$52.5M

ESA; Astroscale; UK Government; Starfish Space press releases

Why now: the rules just changed, and money is moving.

5 yrs

FCC limit to deorbit LEO satellites after their mission ends (for launches after Sept 2024).

0.9

Required disposal success (FCC Part 100, July 2026): up to 1 in 10 satellites may fail and still comply.

$1.1–1.6B

Debris monitoring + removal market in 2025 (analyst estimates); removal alone ≈ $0.15B, growing ~39% a year.

FCC 22-74 (2022); FCC 26-47 (2026); Mordor Intelligence, ResearchAndMarkets, The Business Research Company

Three customers, one route.

Segment	What they need	How they buy	Their challenge
1 · Space agencies ESA, JAXA, UKSA, NASA, US Space Force	Measurable collision-risk reduction per dollar; old rocket bodies gone	Competitive contracts, paid at milestones; want a demo first	Budget cycles; foreign-owned debris needs the owner’s consent
2 · Satellite operators Iridium, Eutelsat OneWeb, Planet, Spire	A plan for satellites that die before they can deorbit; meet the 5-year rule	Per-object service; budget for end-of-life; insurance	Price-sensitive: today’s prices “don’t make sense”
3 · Insurers space underwriters and brokers	A credible way to retire risk and pay claims	Removal bonds or assurance bundled with policies	Won’t commit until removal is demonstrated

III · CUSTOMER SEGMENTS · OPERATORS (PRODUCT A)

“I expect the cost is really in the millions or tens of millions, at which price I know it doesn’t make sense.”

Matt Desch, CEO of Iridium, on paying to remove its dead satellites (2019). He floated $10,000 per satellite.

24

dead first-generation Iridium satellites still in orbit in our Oct 2026 data (662 kg each)

$12.5M

modelled per removal on a 10-stop Iridium route, launch included; a reusable truck pushes lower (≈$5.2M on our first route)

The gap to $10,000 is real. We close part of it now; we’re testing the rest in interviews.

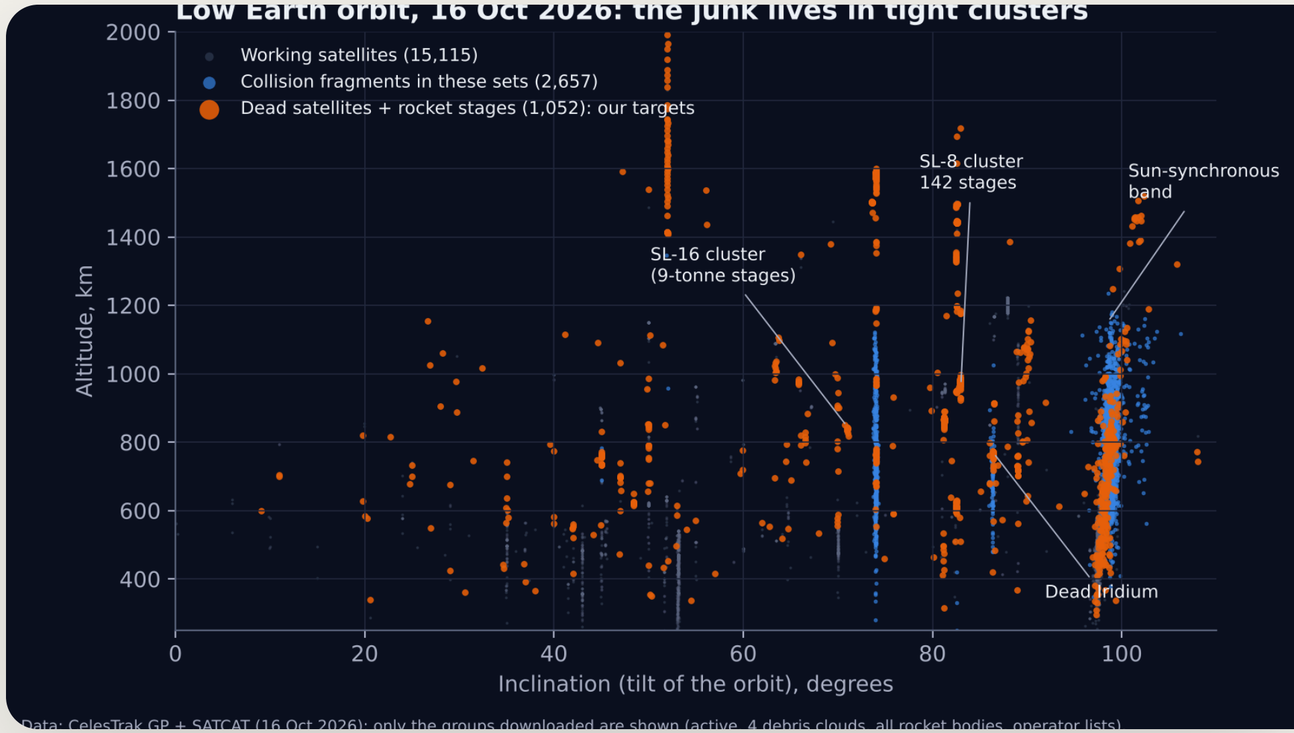
Quote: C. Henry, SpaceNews, 30 Dec 2019 (via ABA Business Law Today, 2022) · CelesTrak SATCAT

Our first route starts where the cheap rockets already go.

26 rocket stages owned by the US, India and France sit in the sun-synchronous band, 650–850 km up. Rideshare launches drop satellites in that band, so our truck reaches its first stop in 80 days.

Why agencies: they already pay for removal, at milestones, and their own countries own these stages.

The bigger prize, later: 142 Soviet SL-8 stages at 83°. They need Russia's consent and a year-long climb, so they cost more: $13.0M each.



Our analysis of the CelesTrak catalogue, 16 Oct 2026; McKnight et al. (2021)

IV · UNIQUE VALUE PROPOSITION

We don’t chase the junk. We let Earth bring it to us.

Space-debris removal priced like a garbage route: one truck, ten stops, a fixed fee per object, paid when it’s confirmed coming down.

IV · UNIQUE VALUE PROPOSITION

A route beats a custom mission.

	One-object missions (today)	Orbital Curbside
Objects per launch	1–2	**10 per route; ~50 per reusable truck**
Cost per object	$40–93M (public contracts)	**$12.2M modelled, launch included; ≈$5.2M with reuse**
How it turns	Burns fuel to change orbit planes	**Waits for Earth’s bulge to line planes up**
Engine	Mostly chemical	**Electric (xenon): ≈200 kg, launch to last stop**
Payment	Custom contract per mission	**Fixed fee per object, paid on confirmed deorbit**
Planning	Hand-designed per target	**Software picks the route from the live catalogue**

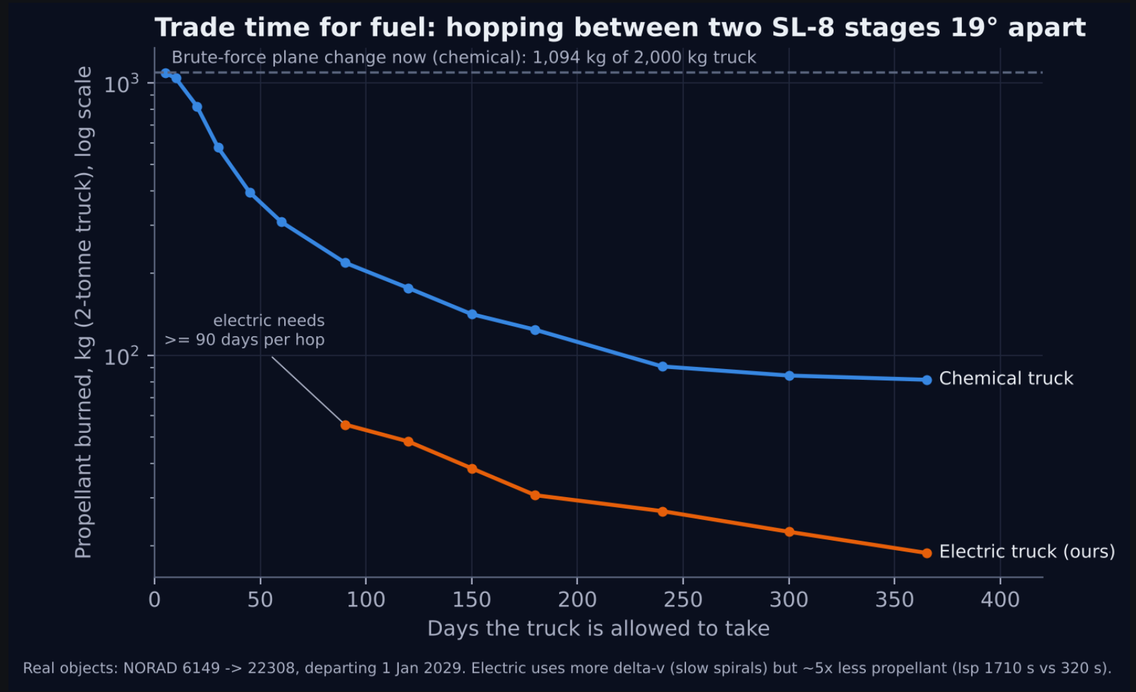
Contract values: slide 6. Our figures: model estimates, base assumptions (appendix).

01 **Park.** Move to a slightly different height than the next target.

02 **Drift.** Earth’s bulge turns the two orbits at different speeds, so the gap closes for free.

03 **Hop.** A short electric burn to meet the target.

04 **Tag.** Attach a small deorbit kit that drops the object into the atmosphere. Drive on.



Closing a 19° plane gap: burn now ≈ 1,094 kg of propellant; drift for a year ≈ 19 kg.

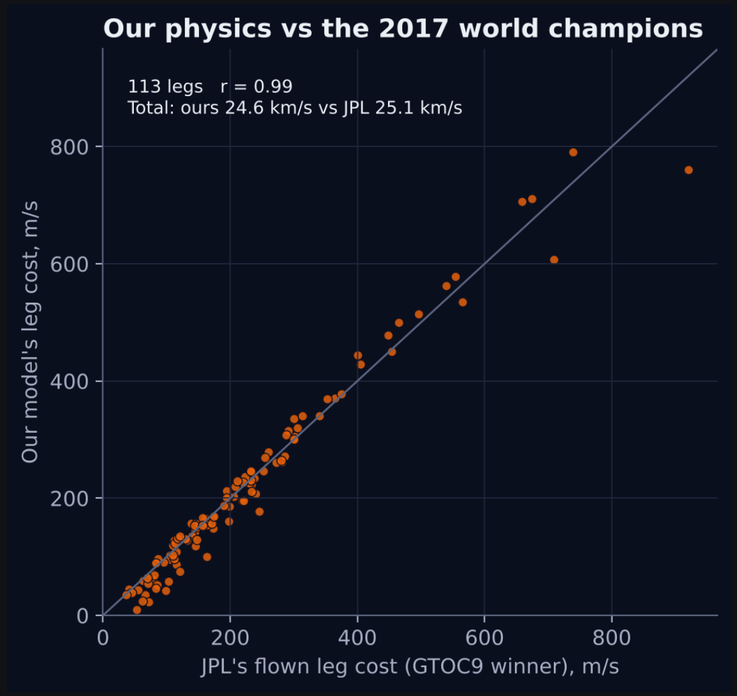
We checked our math against NASA JPL’s winning solution.

2.2%
total Δv difference across JPL’s 113 legs (r = 0.99)

0.06°
median plane error vs SGP4, the model used to track real satellites (0.30° on our first route)

2.6 m/s
median correction per leg in a full-gravity check

2.4×10^{21}
possible routes; four search methods compete to pick one



Each dot is one leg of JPL’s winning GTOC9 solution, recomputed by our model.

Petropoulos et al. (2018), GTOC9 JPL results; Vallado et al. (2006), SGP4



One truck. Ten stages. Launch to last stop in 16 months.

10
stages removed

490
days from launch

198 kg
of xenon, total

Real orbits from the 16 Oct 2026 catalogue. Concept truck sized from our model. Both run offline in a browser.

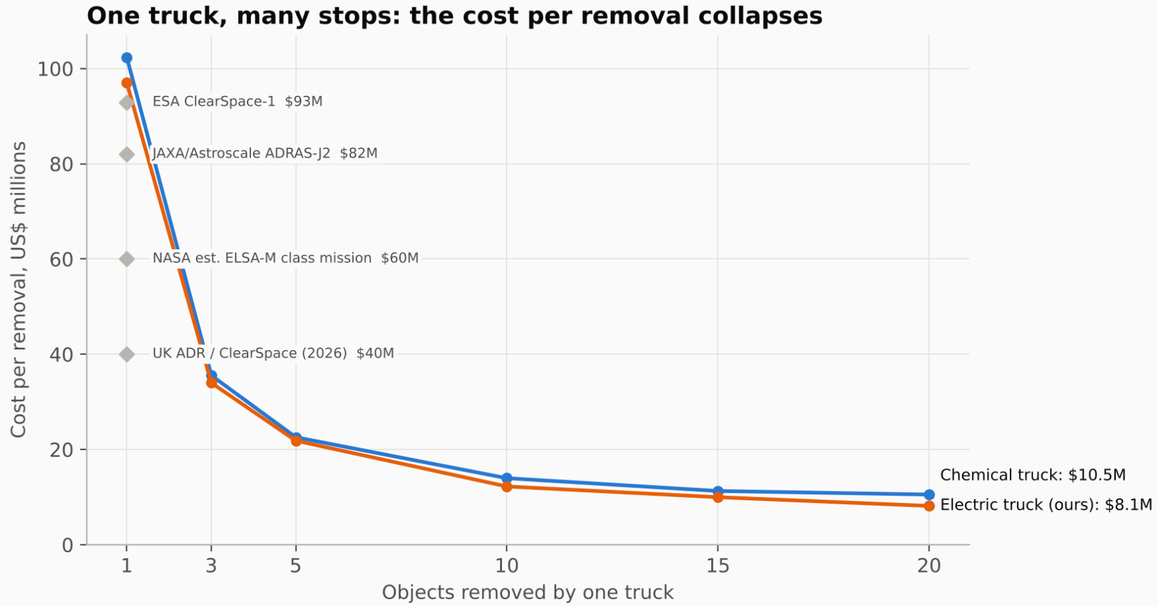
More stops per trip, lower cost per object.

$97M one object per mission, same hardware

$12.2M first route, 10 stops (P10–P90: $11.2–15.4M)

$5.2M reusable truck, ~50 objects over ~6 years

All include a rideshare launch. Price at a 20% margin: $15.3M.
Biggest unknown: truck hardware ($50–115M).



Cost per removal vs stops per route, launch included (our model)

Model estimates; inputs from NASA OTPS (2023), SpaceX rideshare pricing, Busek; assumptions in appendix

What could stop us, and what we do about it.

Capture

Grabbing a tumbling stage hasn't been done yet. We follow ADRAS-J2 and partner rather than invent the arm.

Cost

Truck and kit prices are estimates, and a rideshare this heavy needs a custom quote. We test them with experts.

Legal consent

Owners keep control of their objects. Our first route uses US, Indian and French stages.

Re-entry safety

Heavy objects get a controlled re-entry over the ocean, not a slow fall.

2026–27

Validate: customer and expert interviews; refine the model

2027–28

Watch ADRAS-J2 capture; sign a capture and bus partner

2028–29

Pathfinder: 2–3 stops in the sun-synchronous band

2029+

First 10-stop truck

Outer Space Treaty (1967), Art. VIII · Astroscale ADRAS-J2

VI · CONCLUSION

Removal priced like a route, not a moon shot.

1 Today’s removal costs $40–93M per object; our first route models $12.2M, launch included.

2 A cheap rideshare gets us there; Earth’s bulge does the steering. Math checked against JPL and SGP4.

3 Impact: one reusable truck removes ~50 old stages over ~6 years.

THE ASK

[$__] in pre-seed funding for 12 months: finish customer validation, a kit design study, and a pathfinder partner.

Scan to fly the route

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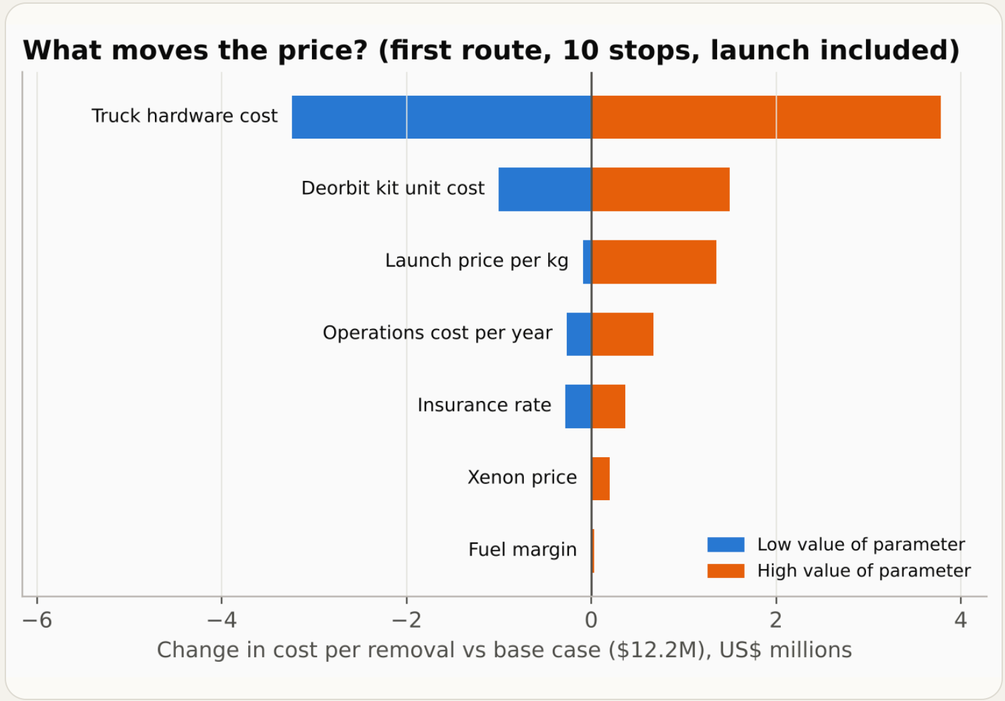
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Full list with links: written paper, references [1]–[28]

Input	Low · Base · High	Source
Truck hardware	$50 · 80 · 115M	NASA OTPS (2023); base = assumption
Deorbit kit, each	$0.5 · 1.5 · 3.0M	Assumption: no public price; testing in interviews
Launch	$7.0 · 7.5 · 15k/kg	SpaceX rideshare (2026–27), to 550 km sun-sync; then the truck climbs
Xenon	$2.8 · 2.8 · 12k/kg	Market reports (2024–26)
Operations	$3 · 5 · 10M/yr	Assumption
Insurance	5 · 8 · 12%	Assumption, of hardware + launch
Δv margin	10 · 10 · 25%	Our JPL and SGP4 checks
Truck	600 kg dry; Hall thrusters 0.202 N, Isp 1,710 s (Busek BHT-1500 class); 10 days per stop	
Kits	Solid motor, Isp 285 s; lower perigee to 250 km; ≈85 kg each on the first route (≈130 kg for a 1.4 t SL-8)	



Swing in cost per removal from each input, others at base. Truck hardware matters most ($9.0–16.0M).

Base case: first route, 10 stops, 490 days from launch, launch included. Sources in paper.